



Making Education in Rural India Affordable

With an inadequate computing infrastructure impeding IT penetration in rural India, Microsoft's MultiPoint SDK 1.1 may provide the much required impetus.

Vandana Sharma

"It lets people learn things they didn't think they could learn before, and so in a sense it is all about potential." These words of Steve Ballmer, CEO, Microsoft Corp, aptly sum up the phenomenal impact that information technology can have on the way we lead our lives. An example of one such technological breakthrough is Microsoft's MultiPoint software development kit (SDK) 1.1.

Developed by Microsoft Research India, MultiPoint allows two to 50 students to collaboratively work on and learn from educational software on one computer, using multiple computer mice that drive specifically-designed on-screen cursors, says Latif Nathani, general manager, Unlimited Potential, Microsoft India.

"This not only offers a more affordable solution—the only increase in cost is that of the additional mice—it also creates an active, collaborative learning experience that engages each and every student," he adds.

The need that drove the innovation

As per a recent research conducted by Microsoft to understand how schools in India use PCs, it was found that only 7.02 per cent of the total government schools have computers and even there, the one or two PCs per class are surrounded by several kids. "In most cases, one observes that a dominant 'bully' hogs the mouse and keyboard. The question was how to multiply the utility of the existing PCs without exponentially increasing the cost. This is how MultiPoint SDK was conceptualised by Microsoft Research India.

"On one hand, the technology enables



collaborative learning between students, and on the other, helps a teacher engage more effectively with students and monitor each child's participation and progress, enabling all students to become active learners, thus enhancing the overall classroom experience," says Nathani.

A platform to build collaborative school content

Using the MultiPoint SDK, curriculum developers can build unique and collaborative applications for schools. It supports new technology platforms like Visual Studio 2008 and includes support for technologies like Flash. The download includes tutorials, white papers and videos to help developers navigate the SDK and build applications.

The actual splitting of screens is completely implemented at the end-user application. Hence, the number of screens that the monitor will be divided into, depends on the screen resolution and end user ap-

plication that is being used.

The reach and the impact

An interesting case study is that of Vidya Pratishthan's Institute of Information Technology (VIIT), which is leveraging the platform to multiply the impact of its initiative to take computing to the rural schools of Maharashtra. VIIT is conducting trials of MultiPoint in its fleet of 30 technology module buses, which it uses to spread computer literacy in rural schools of the region. So far it has conducted pilots in 10 schools.

"VIIT will offer these applications to interested schools from this academic year. VIIT has developed around 10 to 15 applications for Classes 4 to 9, in mathematics, English, biology and chemistry," says Dr Amol Goje, director, VIIT. His observation is that MultiPoint "...is a fantastic technology—simple and yet powerfully apt for a country like India where scarcity of computers for education is a reality." ■